

SIDDHARTH SINHA

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Computer Science undergraduate focused on cybersecurity, digital forensics, and AI-enabled full-stack systems. Builds offline-first investigation and threat-intelligence tools with FastAPI, Python, SQL, JavaScript, NetworkX, D3, and local LLM workflows.

EDUCATION

Amity School of Engineering & Technology, AUH

Haryana, India

B.Tech. in Computer Science Engineering - CGPA: 8.46/10

2023-2027

- Leadership: Mission Veerangana organising team; former Nihon Culture Club Coordinator; member of University Digital Club.

EXPERIENCE

Gurugram Police Cyber Security Summer

Gurugram, India

Internship (GPCSSI)

Cyber Security Summer Intern

Jun 2026

13th edition; 1,000+ students selected from 15,000+ national applicants for a month-long in-person cybersecurity programme.

- Developed ARGUS – Advanced Records & Geospatial Unified Surveillance, completed practical training under the mentorship of Dr. Rakshit Tandon across cyber hygiene, AI scam prevention, OSINT, and digital investigation.
- Worked through investigation-oriented learning modules focused on real-world cybercrime awareness, evidence handling, and defensive-security thinking.

Amity University Gurgaon

Gurugram, India

Student Intern

Sep 2025-Present

Private research university with multidisciplinary academic and professional programmes.

- Manage backend content workflows and structured institutional data across digital platforms, improving accuracy and accessibility.
- Support media operations through event coverage, photography, and standards-aligned content publishing.

National Informatics Centre

New Delhi, India

Cyber Security Intern

Jun 2025-Jul 2025

Government of India ICT organisation under the Ministry of Electronics and Information Technology.

- Performed hands-on vulnerability assessments and security reviews across systems and networks in a government cybersecurity environment.
- Applied threat detection, risk analysis, and mitigation concepts to practical defensive-security scenarios.

Embrizon Technologies

Remote

Cyber Security Intern

Dec 2024-Feb 2025

Technology training and development organisation.

- Completed intensive training in penetration testing, vulnerability assessment, network security, and ethical-hacking workflows.
- Used common assessment tools and frameworks to identify vulnerabilities and understand defensive controls.

Prodigy InfoTech

Remote

Web Development Intern

Sep 2024-Oct 2024

Technology firm providing hands-on software engineering experience.

- Built and refined frontend and backend web features using HTML, CSS, JavaScript, and integrated application workflows.

SELECTED PROJECTS

Project ARGUS | Telecom forensics platform for CDR/IPDR investigation | [GitHub](#)

- Built a full-stack case-management platform for analysing CDR, IPDR, and cell-tower records with FastAPI, SQLAlchemy, Pandas, NetworkX, PostgreSQL/SQLite, D3, Chart.js, and Leaflet.
- Implemented investigator workflows including network graphs, tower movement maps, timelines, service attribution, cross-subject correlation, searchable records, evidence pinning, audit logs, and court-ready dossier export.
- Designed spatiotemporal and network inference modules for risk scoring, impossible travel, SIM-swap/burner signals, hidden links, meetings, beaconing, VPN/proxy use, rare destinations, and data-exfiltration patterns while keeping CDR phone subjects and IPDR IP subjects separate.
- Added offline-first AI insights using local Ollama or a fine-tuned Qwen2.5 LoRA pipeline trained on 3,218 project-specific examples across 15 investigation question types.

Project SENTINEL | AI-powered network threat intelligence and monitoring platform | [GitHub](#)

- Engineered a cybersecurity platform with FastAPI, React, and WebSockets for real-time network discovery, monitoring, and threat analysis.
- Integrated ARP scanning, Nmap enumeration, and local CVE intelligence to identify vulnerable networked devices.
- Built an LLM/NLP control layer with Ollama for command-driven security operations and offline-first SOC-style workflows.

AgentRecon | Modular AI reconnaissance and vulnerability analysis pipeline | [GitHub](#)

- Automated reconnaissance with Nmap, Masscan, Nikto, Gobuster, and SSLyze for asset discovery and vulnerability enumeration.
- Developed CVE enrichment modules using NVD and AlienVault OTX, then applied AI filtering to reduce false positives and improve prioritisation.

AI Research Agent | Private-document RAG and semantic retrieval system | [GitHub](#)

- Built a RAG pipeline for private PDFs using NLTK preprocessing, chunking, hybrid dense/TF-IDF retrieval, and grounded LLM answer generation.
- Trained a Siamese BiLSTM embedding model in Keras to improve semantic similarity matching across document chunks.

F1 Insight Platform | Full-stack racing analytics and strategy simulator | [GitHub](#)

- Developed FastAPI, Streamlit, SQLAlchemy, and XGBoost modules for telemetry analysis, race-outcome prediction, pit-stop strategy, and conversational querying.

Additional: HITMEN moderation tracker, Innovathon event platform, Office of International Affairs AUH website. - <https://global-connect-hub-two.vercel.app/> - <https://innovathon2.onrender.com/> - https://sid1125.github.io/hitmen_epi_prevention/

TECHNICAL SKILLS

- Cybersecurity: vulnerability assessment, penetration-testing fundamentals, threat modelling, OWASP Top 10, CTF practice, Wireshark, Burp Suite, Nmap, Metasploit.
- Programming and backend: Python, C, C++, FastAPI, SQLAlchemy, Pandas, NetworkX, REST APIs, WebSockets, PostgreSQL, SQLite, MySQL, MongoDB basics.
- Frontend and data: HTML, CSS, JavaScript, React basics, Streamlit, D3.js, Chart.js, Leaflet, data visualisation, dashboards.
- AI/ML: Ollama, OpenAI API, RAG, TF-IDF, embeddings, Keras, XGBoost, LoRA fine-tuning concepts, local-first AI workflows.

CERTIFICATIONS, ACHIEVEMENTS & LANGUAGES

- Certifications: IBM Cybersecurity Fundamentals; Cisco Introduction to Cybersecurity; Fortinet Certified Fundamentals; freeCodeCamp Responsive Web Design and JavaScript Algorithms/Data Structures; Google Cloud Computing Fundamentals and Generative AI Study Jam.
- Achievements: 13th of 80 teams in IEEE Computer Society x IIT Bombay Trust Lab CTF; Smart India Hackathon 2024 participant for PS ID SIH1674; 100-day TryHackMe streak, Level [0x8][HACKER].
- Languages: English and Hindi - native/bilingual proficiency; Japanese - elementary proficiency.